EIN 4343 Inventory and Supply Chain Systems

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EIN4343 INVENTORY & SUPPLY CHAIN SYSTEMS

INVENTORY SYSTEMS: PRELIMINARIES

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Basic definitions

- **Inventory** is the stock of any item or resource in an organization.

- A **stock-keeping unit (SKU)** is a specific unit of stock to be controlled in an inventory system.

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Types of inventories

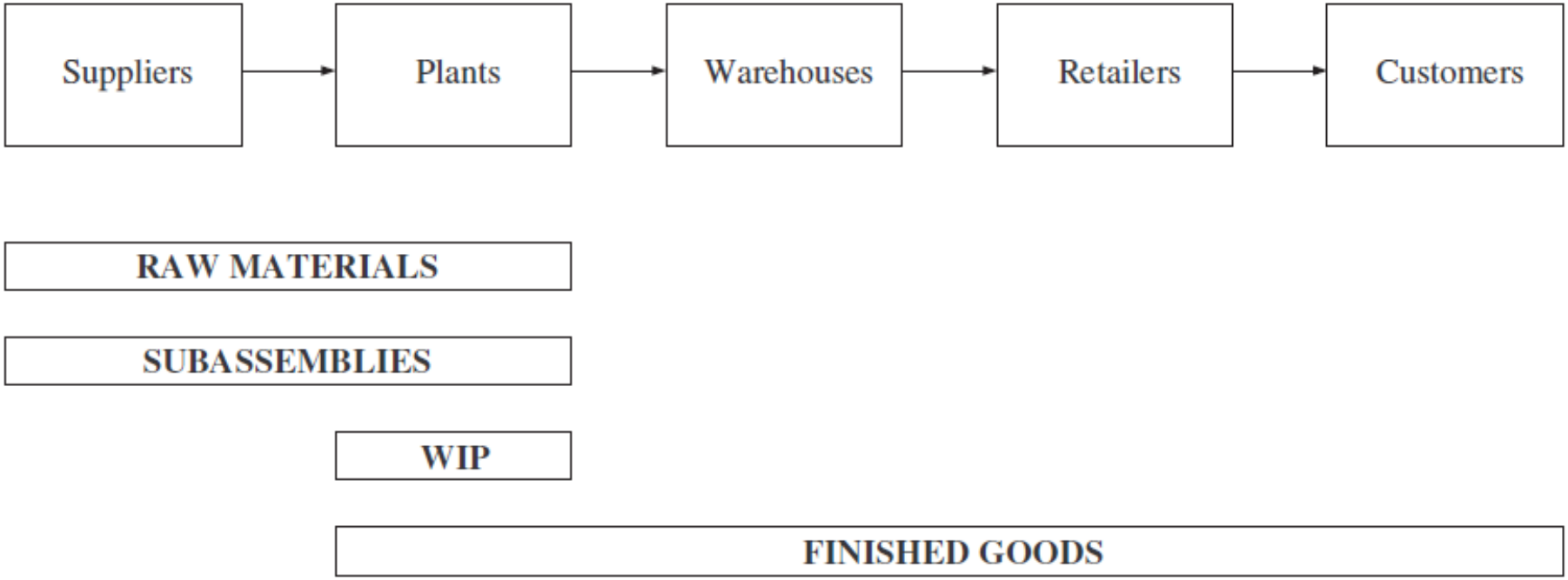

Figure 1: Types of inventories held in a supply chain.

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Need for inventories

1. Economies of scale
2. Uncertainties
3. Speculation
4. Smoothing
5. Transportation
6. Inventory control costs

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1. Economies of scale

When you go to the grocery story how many bananas or how many rolls of toilet paper do you buy?

- When a supplier sets up a production line or a distributor makes a shipment or a customer makes a purchase, there is typically a fixed cost associated with the setup or the shipment or the purchase. Hence, the production or shipment or purchase lot size can be larger than the immediate demand and what is not immediately needed is kept in inventory.

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2. Uncertainties

Do you know exactly how many bananas or how many rolls of toilet paper do you will use next week?

- The demand process is uncertain by nature.
- The supply process can be uncertain too (e.g., yield uncertainty, lead time uncertainty).
- All such uncertainties could lead to a mismatch between the demand and supply processes, leading to a need to keep inventory.

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3. Speculation

When a hurricane is approaching, do you buy more gas? Or, when you go to Publix and see that the beverage of your choice is “buy 3 get 1 free,” do you buy 4?

- If a supply shortage or a price increase is anticipated, then inventory can be kept.

- Southwest Airlines vs. United Airlines

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4. Smoothing

When an exam is upcoming, do you study a little bit everyday or cram the night before?

- When a new generation PlayStation or Apple iPhone is released, the immediate demand can be so high that the supplier's manufacturing and supply chain capacities may not be able to meet the demand.
- In such cases, the supplier starts production before the planned release date and builds inventory to meet the demand at product launch.

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5. Transportation

When do you buy more? When you go shopping with a car or a bike? How long does it take to send a ship from China to the United States?

- Both transportation capacity (as a facilitator) and cost (as a barrier) play a role in shipment of lot sizes.
- Larger shipment sizes lead to larger inventories being kept.

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6. Inventory control costs

How much do you buy, when you buy paper clips or Q-tips©?

- For some items, keeping track of the amount in inventory can cost more than keeping the inventory. Hence, if the item is low cost (and also typically low space/weight), it can be purchased in bulk and kept in inventory.

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Relevant costs for inventories

- Procurement/production cost
  - What you pay to procure/produce the item
- Inventory holding cost
  - What you incur for items in inventory
- Inventory shortage costs
  - What you incur for inventory shortages
  - Depends on how shortages are handled: Backordering, lost sales, or substitution

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Inventory management questions

The fundamental questions that need to be answered are the following:

1. What to stock?
2. How often to review (i.e., determine) the inventory status?
3. When to order?
4. How much to order?

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Inventory model characteristics 1

I. Planning timeline

- Infinite vs. finite horizon
  - We can assume that the planning horizon is infinite, i.e., beginning and end do not exist, or we can assume that the planning horizon is finite, i.e., beginning and end are known.

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Inventory model characteristics 2

II. Demand process

- Deterministic vs. stochastic process
  - We may assume that the demand is known with certainty (i.e., deterministic), or demand is uncertain (i.e., stochastic).

- Stationary vs. non-stationary process
  - We may assume that the parameters of the demand do not change over time (i.e., stationary), or the parameters changes over time (i.e., non-stationary).

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Inventory model characteristics 3

III. Lead time process

- *Lead time* is the duration that elapses between the time when the procurement/production order is placed and when the order is received.

- Deterministic vs. stochastic process
  - We may assume that the lead time is known with certainty (i.e., deterministic), or lead time is uncertain (i.e., stochastic).

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Inventory model characteristics

IV. Inventory review scheme

- *Inventory review scheme* dictates how frequently the inventory status is checked.

- Continuous vs. periodic review
  - We may assume that the on-hand inventory level is checked and known all the time (i.e., continuous), or inventory level is checked and known periodically (i.e., periodic).

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Inventory model characteristics 4

V. Handling of excess demand

- Excess demand occurs when the inventory on hand is not sufficient to satisfy the demand received.

- Backlogging, lost sales, and substitution
  - We may assume that the excess demand is backlogged, if the customers are willing to wait until the next replenishment cycle when the inventory is not sufficient.
  - We may assume that the excess demand is lost, if the customers are not willing to wait.
  - We may assume that the excess demand is substituted, if the customers are willing to substitute (upward or downward) when there is not sufficient inventory.

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Inventory model characteristics 5

VI. Inventory Life

- Perishable vs. unperishable
  - We can assume that the item is perishable and cannot be kept in inventory beyond a certain time (e.g., unfrozen meat, dairy, blood), or we can assume that the item is non-perishable (e.g., a Twinkie).

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Inventory model characteristics: Epilogue

- Why should you care?

- These model characteristics are important to fully conceptualize an inventory system.

- You can categorize and model ANY inventory system by relying on these 6 fundamental characteristics.

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EIN4343 INVENTORY & SUPPLY CHAIN SYSTEMS

INVENTORY SYSTEMS: ABC ANALYSIS

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Vilfredo Pareto (1858-1923)

- Italian engineer, sociologist, economist, and philosopher

- Observed that 80% of the land in Italy was owned by 20% of the population

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So-called 80-20 rule

- The most frequently used 20% of the words account for 80% of the word occurrences.
- The richest 20% of the humans have 80% of the world's income.
- In the US, 20% of the patients use 80% of the health care resources.
- 20% of the customers are responsible for 80% of the profit earned.
- 20% of the complaints received come from 80% of the customers.
- People wear around 20% of their clothes 80% of the times.

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Inventory management too?

- Does the 80-20 rule apply in the context of inventory systems too?

- How?

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So-called 80-20 rule

- The most frequently used 20% of the words account for 80% of the word occurrences.

usage

- The richest 20% of the humans have 80% of the world's income.

value

- In the US, 20% of the patients use 80% of the health care resources.

usage

- 20% of the customers are responsible for 80% of the profit earned.

value

- 20% of the complaints received come from 80% of the customers.

usage

- People wear around 20% of their clothes 80% of the times.

usage

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Inventory management too?

- Does the 80-20 rule apply in the context of inventory systems too?

- How?
  - By the unit price?
  - By the usage volume?

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Distribution value analysis (DVA)

- Typically, somewhere on the order of 20% of the SKUs account for 80% of the total dollar value of the inventory transactions.

- Some items are more important in terms of the money invested, profit potential, sales or demand volume.

- A Distribution Value Analysis (DVA) curve can be used to classify SKUs to identify the most important, considerably important, and least important items.

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Distribution value analysis (DVA)

1. For each SKU, (a) identify its unit cost and (annual or over a planning period) demand and (b) use the product of these two values to identify the total (annual) dollar contribution value of the SKU.

2. Rank SKUs in their non-increasing order of total annual dollar contribution values

3. Calculate and plot the cumulative percentage of total annual dollar contribution values for the SKUs

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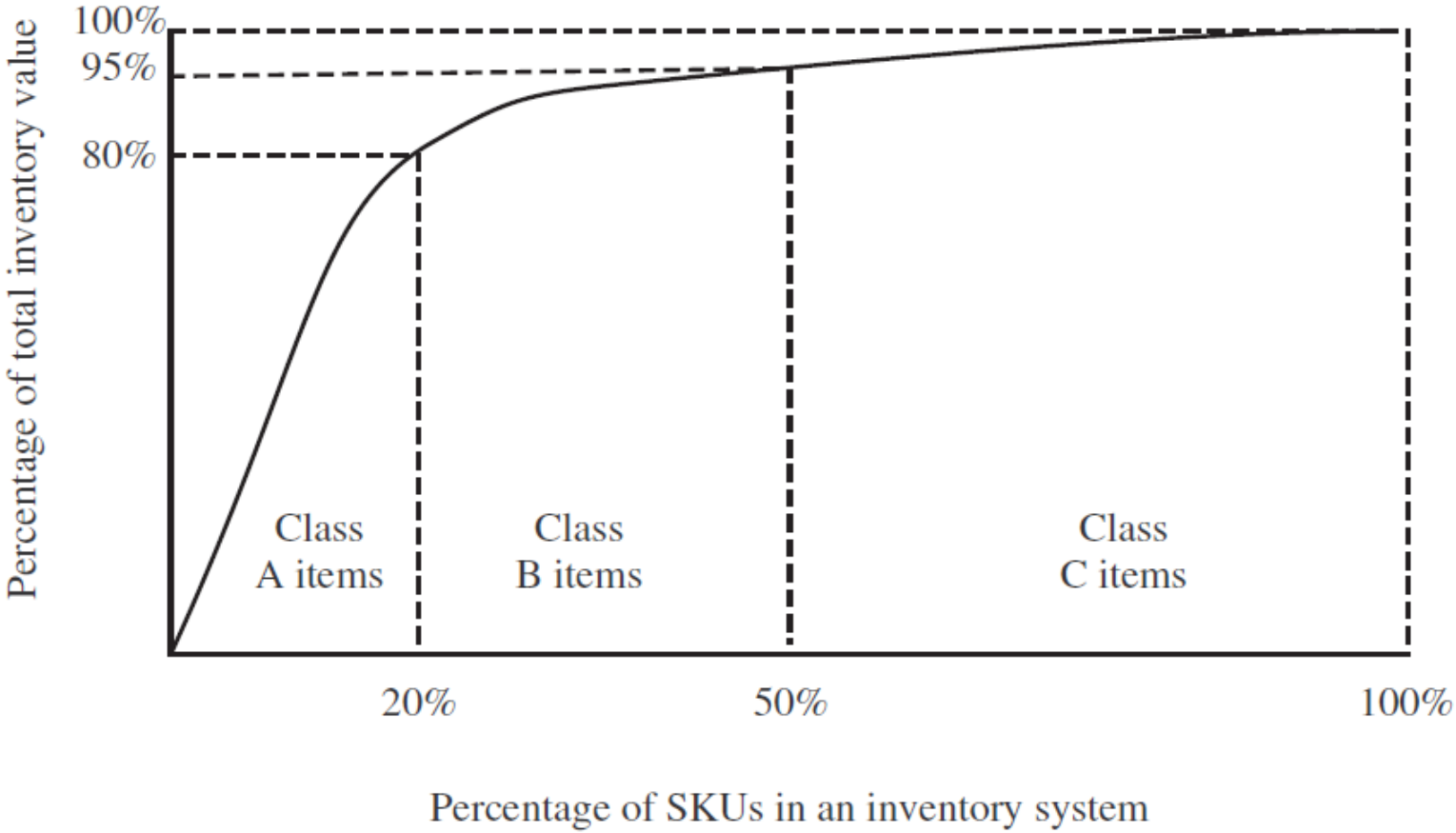
DVA and ABC Classification



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ABC Classification

- Class A items are the most important items in an inventory system which should be monitored closely.
  - Sophisticated inventory control approaches
- Class B items are considerably important items which should be monitored on occasion.
  - Structurally simple and easily implementable inventory control approaches
- Class C items are the least important items in an inventory system which should require minimal tracking.
  - Making bulk purchases

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Example

- What percentage of the total dollar contribution is accounted for by the top 20% of the items? How about the bottom 50% of the items?

Item i	Demand (λ_i)	Cost (c_i)
1	80	422.23
2	514	54.07
3	19	0.65
4	2,442	16.11
5	6,289	4.64
6	128	0.63
7	1,541	2.96
8	4	22.05
9	25	5.01
10	2,232	2.48
11	2	4.78
12	1	38.03
13	6	9.01
14	12	25.89
15	101	59.5
16	715	20.78
17	1	2.93
18	35	19.52
19	1	28.88
20	4	29.86

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Example

Item i	Demand (λ_i)	Cost (c_i)	Total Dollar Value ($\lambda_i c_i$)	% Value	Cumulative % Value
4	2,442	16.11	39,340.62	24.19%	24.19%
1	80	422.23	33,778.4	20.77%	44.97%
5	6,289	4.64	29180.96	17.95%	62.91%
2	514	54.07	27,791.98	17.09%	80.00%
16	715	20.78	14,857.7	9.14%	89.14%
15	101	59.5	6,009.5	3.70%	92.84%
10	2,232	2.48	5,535.36	3.40%	96.24%
7	1,541	2.96	4,561.36	2.81%	99.04%
18	35	19.52	683.2	0.42%	99.46%
14	12	25.89	310.68	0.19%	99.66%
9	25	5.01	125.25	0.08%	99.73%
20	4	29.86	119.44	0.07%	99.81%
8	4	22.05	88.2	0.05%	99.86%
6	128	0.63	80.64	0.05%	99.91%
13	6	9.01	54.06	0.03%	99.94%
12	1	38.03	38.03	0.02%	99.97%
19	1	28.88	28.88	0.02%	99.98%
3	19	0.65	12.35	0.01%	99.99%
11	2	4.78	9.56	0.01%	100.00%
17	1	2.93	2.93	0.00%	100.00%
			162,609.1		

- Use 80% and 95% as cutoffs
  - A class: 4, 1, 5, and 2
  - B class: 16 and 15
  - C class: all others

- **Note:** *These cutoff values are not set in stone and human judgement should be used for classification.*

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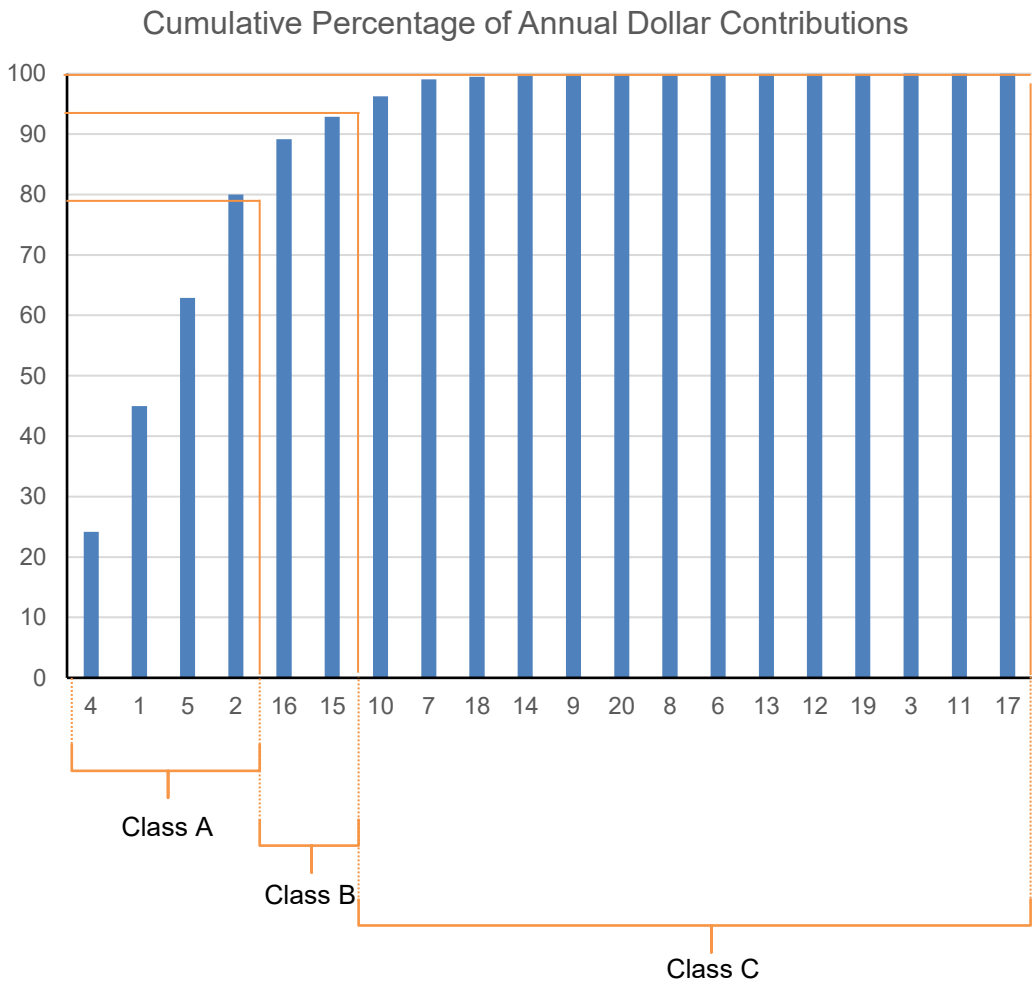
Plot of the cumulative percentages



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Example

Item i	Demand (λ_i)	Cost (c_i)	Total Dollar Value ($\lambda_i c_i$)	% Value	Cumulative % Value
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16	715	20.78	14,857.7	9.14%	89.14%
15	101	59.5	6,009.5	3.70%	92.84%
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6	128	0.63	80.64	0.05%	99.91%
13	6	9.01	54.06	0.03%	99.94%
12	1	38.03	38.03	0.02%	99.97%
19	1	28.88	28.88	0.02%	99.98%
3	19	0.65	12.35	0.01%	99.99%
11	2	4.78	9.56	0.01%	100.00%
17	1	2.93	2.93	0.00%	100.00%
			162,609.1		

- Use 90% and 99.1% as cutoffs
  - A class: 4, 1, 5, and 16
  - B class: 15, 10, and 7
  - C class: all others

- **Note:** *Use your judgement!*
- **One Final Note:** *Look at C items carefully!*

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Epilogue

- Simple procedure: Some arithmetic and sorting

- Pay attention
  1. *Cut-off points for Class A and B items*: Should they be adjusted based on judgment?
  2. *Implied classification of each item*: Should any item be reclassified based on some relevant institutional knowledge?

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EIN4343 INVENTORY & SUPPLY CHAIN SYSTEMS

INVENTORY SYSTEMS: XYZ ANALYSIS

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What is ABC Analysis missing?

- Demand or usage for an SKU is considered as a cumulative period over a period

- Suppose for an SKU, the annual demand is 120 units
  - Demand can be 10 units per month
  - Demand can be 120 units in the 3rd month

- What is different?

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What does XYZ analysis add?

- XYZ analysis focuses on the standard deviation of usage/demand
  - XYZ analysis is complementary to ABC analysis

- Helps identify which SKUs have steady usage/demand and which SKUs have unpredictable usage/demand

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Standard deviation calculation

1. For each SKU, identify the daily, weekly, monthly or quarterly order sizes and calculate the associated sample standard deviation

2. Rank SKUs in their non-increasing order of their sample standard deviations

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XYZ Classification

- Class X items exhibit little demand variability.
  - Demand does not change as often or as drastically and can be forecasted accurately.
- Class Y items have fluctuating demand.
  - Demand is subject to some variability due to a known or predictable cause and can be forecasted to some degree.
- Class Z items exhibit high demand variability.
  - Demand is sporadic and highly unpredictable due to unknown causes and can be difficult to forecast.

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XYZ Classification

- Class X items are suitable for automated replenishment mechanisms.
- Class Y items are suitable for automated replenishment mechanisms with manual intervention.
- Class Z items are suitable for buy to order type replenishment mechanisms.

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INVENTORY SYSTEMS: ABC AND XYZ

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XYZ complements ABC

- ABC and XYZ analyses can be used together for a more complete picture of managing inventory.

- While one shows which SKUs are important in terms of value, the other one shows how variable the demand processes for the SKUs are.

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Pairing ABC and XYZ

	X	Y	Z
A	High importance, Even demand, Reliable forecasts	High importance, Predictably variable demand, Less reliable forecasts	High importance, Sporadic variable demand, Unreliable forecasts
B	Medium importance, Even demand, Reliable forecasts	Medium importance, Predictably variable demand, Less reliable forecasts	Medium importance, Sporadic variable demand, Unreliable forecasts
C	Low importance, Even demand, Reliable forecasts	Low importance, Predictably variable demand, Less reliable forecasts	Low importance, Sporadic variable demand, Unreliable forecasts

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Epilogue

- ABC and XYZ are about recognizing patterns in usage and demand variability

- But there are always those that do not fit a class
  - An SKU sold only once a year can be A…
  - A “new” SKU may have no historical data…
  - An SKU may require a “life-time buy”...

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